MAIE 6000C: Engineering AI Products: From Prototype to Production

Week 1 - From Prototype to Production

Program: MSc in AI & Entrepreneurship

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MAIE 6000C — Engineering AI Products: From Prototype to Production

Today’s focus:

- Why AI prototypes fail after the demo
- What “production-oriented” means in this course
- What every team project must eventually show
- How the starter repository supports the course workflow

Question

“What is a project you’ve seen that looked impressive in a demo but probably wouldn’t survive real use?”

A demo is not a system

A prototype often has:

- A narrow happy path
- Hidden setup assumptions
- Little or no durable state
- Weak error handling
- Weak testing
- No operational visibility
- No clear handover path

A system needs:

- Persistent workflow state
- Clear interfaces
- Failure handling
- Reproducible startup
- Tests or verification
- Logs, health checks, and documentation

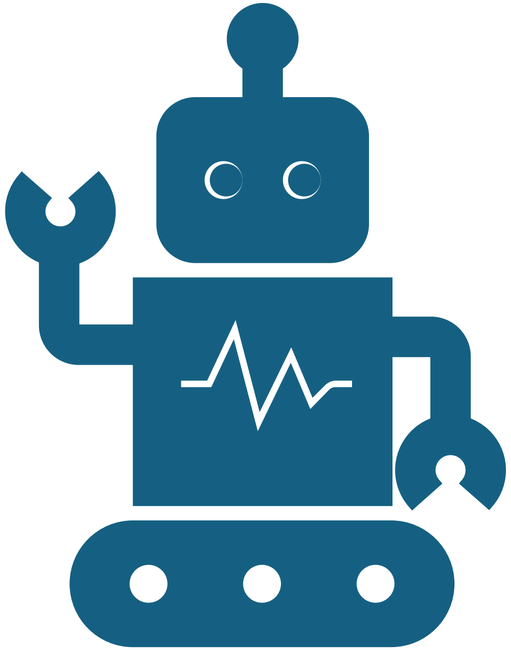
Why AI prototypes fail in real settings

Common failure reasons:

- The model is disconnected from the actual workflow
- Data is not stored or versioned properly
- Users cannot review or correct outputs
- Long-running work blocks the user
- Failures are invisible
- Setup only works on one machine
- Nobody can explain the trade-offs later

Question

Suppose your AI gives a wrong answer. What should happen next in a well-designed system?



Famous AI demo-to-reality failures

McDonald’s + IBM AI drive-thru

- AI ordering was tested in more than 100 restaurants
- The test was ended after real-world accuracy and voice-understanding challenges
- Shows that a demo must work across noise, accents, dialects, menu complexity, and speed pressure

Image source: StraightArrowNews

McDonald’s questions

Why might a system that works at 90% accuracy in a demo still fail in a drive-thru?

What metrics would matter more than raw model accuracy here?

Image source: oecd.ai

Famous AI demo-to-reality failures

Zillow Offers

- Automated home-pricing and buying decisions contributed to major financial losses
- The company shut down the home-buying program
- Shows that prediction systems are risky when they directly control expensive real-world decisions

Zillow question

When AI is connected to expensive decisions, what extra controls are needed beyond accuracy?

Famous AI demo-to-reality failures

Google Bard launch demo

- The chatbot gave an incorrect factual answer in a public launch example
- The mistake received major attention because it happened during a high-profile AI rollout
- Shows that fluent AI responses can still be wrong and need verification

Image source: firstpost.com

Bard question

Why are fluent wrong answers more dangerous than obviously broken answers?

Image source: bbc.com

Famous AI demo-to-reality failures

- **Air Canada chatbot**
  - A customer received incorrect policy information from the airline's chatbot
  - The company was held responsible for the chatbot's answer
  - Shows that AI support tools need guardrails, escalation paths, and clear accountability

Air Canada question

If a chatbot gives official-looking advice, who is accountable?

What this course rewards

This course rewards systems that are:

- Coherent
- Well scoped
- Reproducible
- Reviewable
- Tested or credibly verified
- Operationally visible
- Technically honest
- Maintainable enough to hand over

It does not primarily reward:

- Model novelty
- Big but fragile scope
- Flashy UI without backend substance
- Notebook-only workflows

The AI component is not the whole project

In this course, the AI component is **one workflow step.**

A complete project also needs:

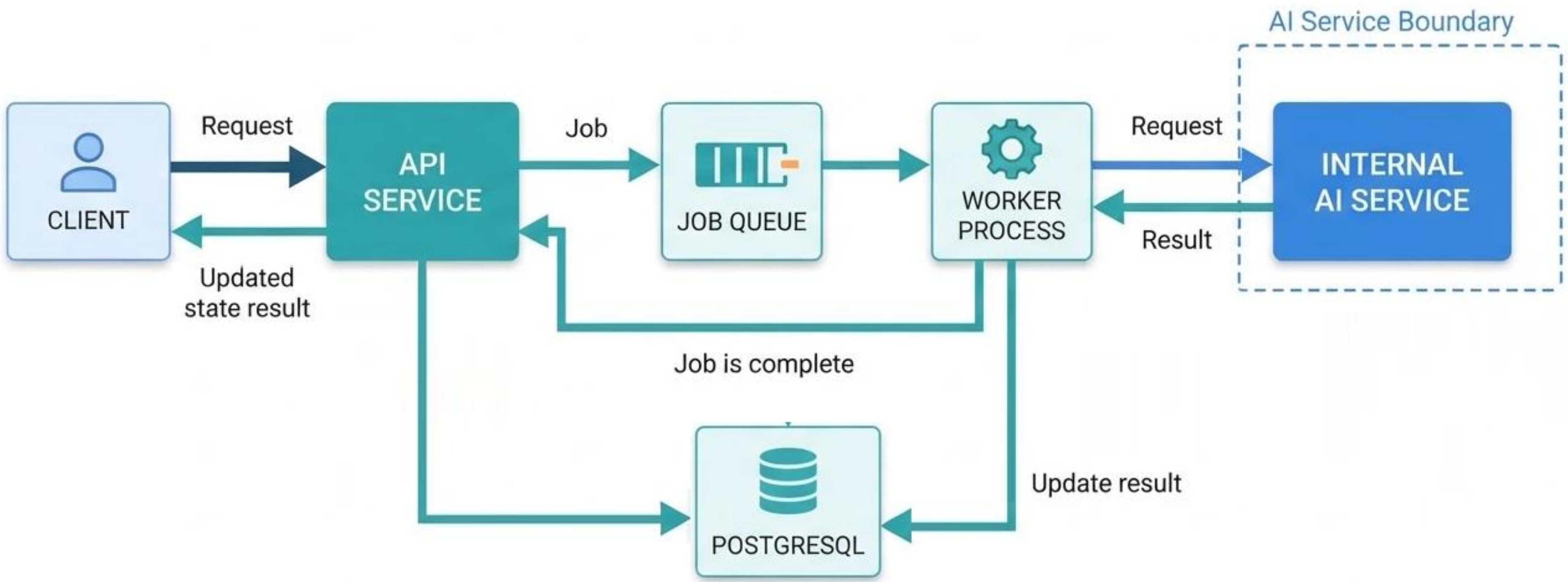
- User input or system input
- Persistent data
- API or service layer
- Async / worker path
- AI-enabled function
- Human review or fallback path where appropriate
- Output, status, or decision record
- Operational signals

Course system contract

Every passing team project must include:

- Relational persistence
- API or service layer
- Worker or async path
- Integrated AI-enabled function
- Reproducible local startup
- Tests or credible verification
- Documentation
- Structured logs
- Health checks
- Basic observability

Starter repository architecture



Weak vs Strong project direction

Weak direction	Strong direction
• “AI platform for operations” • “Smart dashboard” • “Chatbot for everything” • “Predict anything from all data”	• **One** clear user • **One** concrete workflow • **Clear** input and output • **Persistent** state • **Worker path** • **Bounded** AI function • **Human review** or fallback

Start with **one workflow** and **one vertical slice**.

Today’s lab task

In the project lab today, you will:

- Review the project brief pack.
- Inspect the starter repository structure.
- Identify one or two possible project directions.
- For each direction, answer:
  - Who is the user?
  - What enters the system?
  - What is persisted?
  - What happens asynchronously?
  - Where does AI help?
  - What is the smallest Week 7 demo?